9 **(a)** For each application of radioactive isotopes, state and explain which type of radioactive emission is suitable and suggest an appropriate half-life for the isotope.

(i) household smoke alarm

type of radioactive emission .....

explanation .....

.....

half-life .....

[3]

(ii) measuring the thickness of aluminium strips produced in a factory

type of radioactive emission .....

explanation .....

.....

half-life .....

[3]

(b) Lead-208 ($^{208}_{82}Pb$) has the highest nucleon number of the stable isotopes of lead.

Explain why lead-214 ($^{214}_{82}Pb$) is radioactive.

.....

.....

.....

..... [2]

(c) State **two** different sources of background radiation.

1 .....

2 .....

[2]

[Total: 10]

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